Electrical Properties of Materials

Department of Physics, SCE

1 Semiconductors

Semiconductors have their conductivity between those of insulators and conductors. The typical energy band gap is $1eV$. There are two types of semiconductors: intrinsic and extrinsic.

1.1 Intrinsic semiconductors

Intrinsic semiconductors are pure semiconductors. In an intrinsic semiconductor, all the valence band electrons are filled at absolute zero temperature, that is, $T = 0$ K. At higher temperatures, a few valence electrons jump to the conduction band leaving behind a hole in the valence band. Hence, for every electron created in the conduction band, there is a hole in the valence band. If the hole density (number of holes per unit volume) is N_h and the number of electrons in N_e, then we have

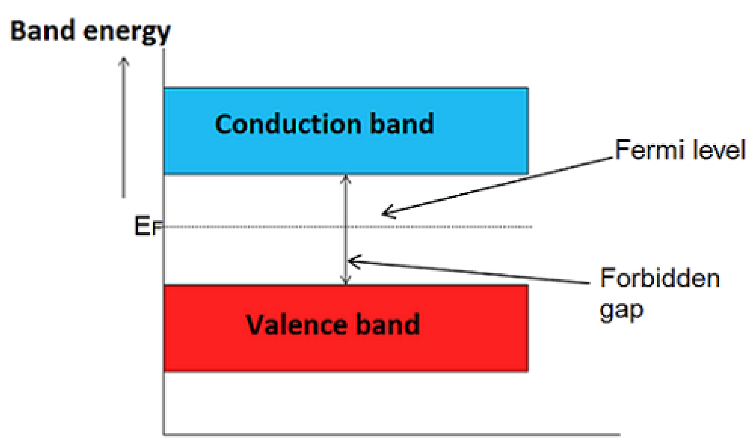
$$N_e = N_h = n_i \quad \text{in intrinsic semiconductors}$$

where n_i is called the intrinsic carrier concentration.

2 Fermi level in intrinsic semiconductors

In intrinsic semiconductors (pure), at $T = 0K$ all the energy levels in the valence band are occupied and all the conduction band energy levels are vacant. When the temperature increases, $T > 0$ K, a few electrons from the valence band are excited and occupy energy levels close to the bottom of the conduction band. The electrons return to the vacant sites in the valence band and again are excited to the conduction band. This is a continuous process due to the thermal energy available at $T > 0$ K. These conduction electrons participate in the electrical conduction. Hence, the conduction process is due to the energy at the top of the valence band and the bottom of the conduction band.

The Fermi level is the average energy level of the bottom of the conduction band and the top of the valence band, that is $E_F = (E_C - E_V)/2 = E_g/2$



3 Fermi level in extrinsic semiconductors (n-type and p-type)

There are two types of extrinsic semiconductors: n-type and p-type.

n-type:

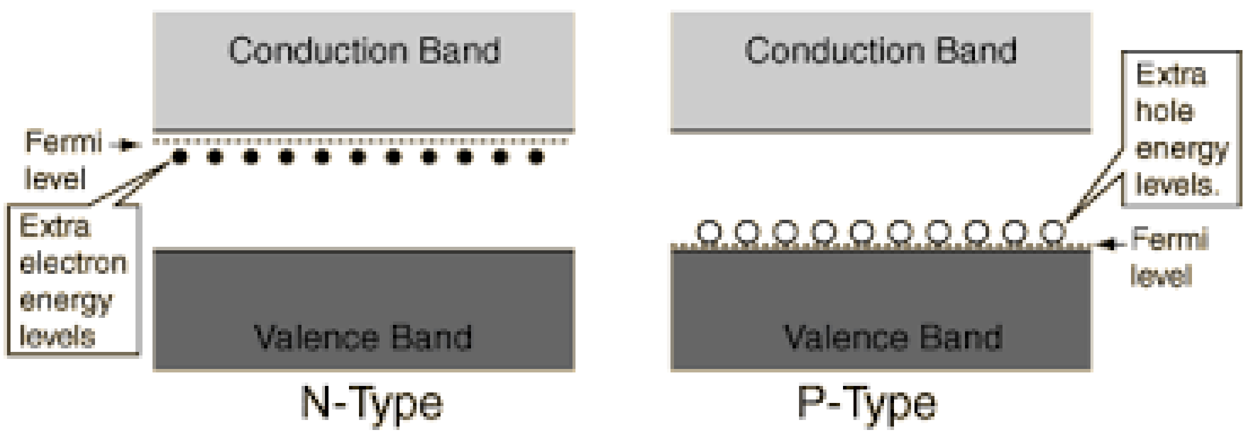
If an intrinsic semiconductor is doped with donor impurities, there will be extra electrons which cannot form covalent bonds with the atoms of the intrinsic material. This type of semiconductors are known as **n-type semiconductors:** These electrons have high energies and occupy energy levels close to the bottom of the conduction band. These energy levels are called donor levels. For temperatures, $T > 0$ K, a few electrons jump from the donor levels to the energy levels close to the bottom of the conduction band. The electrons return to the donor levels. This process of excitation-deexcitation continues. The process of electrical conduction is due to the donor levels and the occupied energy levels in the conduction band. Hence, the **Fermi level for the n-type semiconductors:** is the average of the donor levels and the bottom of the conduction band. Denoting the donor levels by E_d and the bottom of the conduction band by E_c, the Fermi level is approximately given by,

$$E_F = \frac{E_d + E_c}{2}$$

A more accurate calculation shows the Fermi level to be temperature dependent and is given by,

$$E_F = E_c - kT \ln \frac{N_c}{N_d}$$

where N_c is the effective density of states at the conduction band and N_d is the donor number density. **p-type:**



If an intrinsic semiconductor is doped with acceptor impurities, there will be extra holes which cannot form covalent bonds with the atoms of the intrinsic material. This type of semiconductors are known as **p-type semiconductors:** These have energies slightly above the top of the valence band and occupy energy levels close to the top of the valence band. These energy levels are called acceptor levels. For temperatures, $T > 0$ K, a few electrons jump from the valence levels to the acceptor levels. The electrons return to the valence levels. This process of excitation-deexcitation continues. The process of electrical conduction is due to the acceptor levels and the occupied energy levels in the valence band. Hence, the **Fermi level for the p-type semiconductors:** is the average of the acceptor levels and the top of the valence band. Denoting the acceptor levels by E_a and the density of states at top of the valence band by E_v, the Fermi level is approximately given by,

$$E_F = \frac{E_a + E_v}{2}$$

A more accurate calculation shows the Fermi level to be temperature dependent and is given by,

$$E_F = E_v + kT \ln \frac{N_v}{N_a}$$

where N_a is the acceptor number density.

4 Law of mass action

The expression for electron concentration at temperature T is given by,

$$N_e = \frac{4\sqrt{2}}{h^3}\left(\pi m_e^* kT\right)^{3/2} \exp\left(\frac{E_F - E_g}{kT}\right)$$

and the expression for hole concentration at temperature T is given by,

$$N_h = \frac{4\sqrt{2}}{h^3}\left(\pi m_h^* kT\right)^{3/2} \exp\left(\frac{E_F}{kT}\right)$$

where

- m_e^* = effective mass of electron
- m_h^* = effective mass of hole
- h = Planck's constant
- k = Boltzmann constant
- E_F = Fermi energy
- E_g = band gap energy, $E_g = E_C - E_V$, where E_C and E_V are the energies of the bottom of the conduction band and the top of the valence band

Multiplying the two equations gives,

$$\begin{aligned} N_e N_h &= \frac{4\sqrt{2}}{h^3}\left(\pi m_e^* kT\right)^{3/2} \exp\left(\frac{E_F - E_g}{kT}\right) \times \frac{4\sqrt{2}}{h^3}\left(\pi m_h^* kT\right)^{3/2} \exp\left(\frac{E_F}{kT}\right) \\ &= \frac{32}{h^6}\left(\frac{\pi kT}{3}\right)^3 (m_e^* m_h^*)^{3/2} \exp\left(-\frac{E_g}{kT}\right) \end{aligned}$$

The product $N_e N_h$ does not depend on E_F, but depends only on the temperature T. Since, $N_e = N_h = n_i$ for an inrinsic semiconductor,

$$n_i^2 = N_e N_h = \text{ constant for a given T}$$

The constancy of n_i^2 is called the **law of mass action**.

The law of mass action for any semiconductor material, intrinsic or extrinsic, states that the product of the charge carrier concentration remains a constant at any given temperature, even if the doping is varied.

5 Relation between Fermi energy and band gap in intrinsic semiconductors

The band gap in an intrinsic semiconductor is given by $E_g = E_C - E_V$. The inrinsic electron and the hole carrier densities are given by,

$$\begin{aligned} N_e &= \frac{4\sqrt{2}}{h^3}\left(\pi m_e^* kT\right)^{3/2} \exp\left(\frac{E_F - E_g}{kT}\right) \\ N_h &= \frac{4\sqrt{2}}{h^3}\left(\pi m_h^* kT\right)^{3/2} \exp\left(-\frac{E_F}{kT}\right) \end{aligned}$$

Since, $N_e = N_h$ in an intrinsic semiconductor, we have

$$\frac{4\sqrt{2}}{h^3}\left(\pi m_e^* kT\right)^{3/2} \exp\left(\frac{E_F - E_g}{kT}\right) = \frac{4\sqrt{2}}{h^3}\left(\pi m_h^* kT\right)^{3/2} \exp\left(-\frac{E_F}{kT}\right)$$

Cancelling the common terms we get,

$$\begin{aligned}
(m_e^*)^{3/2} \exp\left(\frac{E_F - E_g}{kT}\right) &= (m_h^*)^{3/2} \exp\left(-\frac{E_F}{kT}\right) \\
(m_e^*)^{3/2} \exp\left(\frac{2E_F - E_g}{kT}\right) &= (m_h^*)^{3/2} \\
\exp\left(\frac{2E_F - E_g}{kT}\right) &= \left(\frac{m_h^*}{m_e^*}\right)^{3/2}
\end{aligned}$$

Taking log on both sides,

$$\begin{aligned}
\frac{2E_F - E_g}{kT} &= \frac{3}{2} \ln \frac{m_h^*}{m_e^*} \\
2E_F - E_g &= \frac{3}{2} kT \ln \frac{m_h^*}{m_e^*} \\
2E_F &= E_g + \frac{3}{2} kT \ln \frac{m_h^*}{m_e^*} \\
E_F &= \frac{E_g}{2} + \frac{3}{4} kT \ln \frac{m_h^*}{m_e^*}
\end{aligned}$$

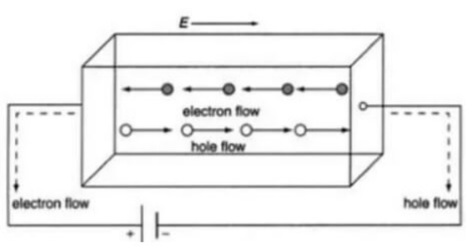
This is the relation between the Fermi energy E_F and the bang gap energy E_g in an intrinsic semiconductor. If the effective masses of the electron and hole are equal, $m_e^* = m_h^*$, then

$$\begin{aligned}
\ln \frac{m_h^*}{m_e^*} &= \ln 1 = 0 \\
E_F &= \frac{E_g}{2}
\end{aligned}$$

that is, the Fermi energy is exactly midpoint between the top of the valence nabd and the bottom of the conduction band.

6 Electrical conductivity of a semiconductor

Consider a material of uniform cross section A and length L. Let the electrical potential difference across it be V. Let the electrical field inside the material be E. Let the free electron density, that is, the number of free electrons per unit volume be N_e. In the absence of an electrical field, there will be random motion of electrons. Electrical current is defined as the net rate of flow of charges in a certain direction, Q/t. Under the action of the electric field



there will be a net average velocity v_e against the electric field. In this case, we need to calculate the total amount of negative charges through the left face in a given time. The number of charges in the bar is given by the number density multiplied by the volume LA, N_eLA. The total charge contained is $Q = N_eLAe$, where e is the electric charge. The time taken by all the electrons in the bar is

$$t = \frac{L}{v_e}$$

Hence, the current, I is given by

$$I = \frac{Q}{t} = \frac{N_eLAe}{L/v_e} = eN_eAv_e$$

The current density J is the current per unit area,

$$J = \frac{I}{A} = eN_e v_e \tag{1}$$

The velocity v_e is proportional to the applied electric field, E,

$$\begin{aligned} v_e &\propto E \\ v_e &= \mu_e E \end{aligned}$$

where, μ_e is called the electron mobility. Hence, J becomes

$$J = eN_e \mu_e E \tag{2}$$

The equation can be rewritten as

$$J = \sigma E \tag{3}$$

, where

$$\sigma_e = eN_e \mu_e$$

is called the electrical conductivity. Similarly, one can obtain the expression for conductivity due to the holes as,

$$\sigma_h = eN_h \mu_h$$

The total conductivity, due to both the electrons and holes is

$$\begin{aligned} J &= eN_e \mu_e + eN_h \mu_h \\ &= e(N_e \mu_e + N_h \mu_h) \end{aligned}$$

In the case of intrinsic semiconductors, $N_e = N_h = n_i$**,**

$$J = en_i(\mu_e + \mu_h)$$

7 Four probe method

A four point probe is a simple apparatus for measuring the resistivity of semiconductor samples. By passing a current through two outer probes and measuring the voltage through the inner probes allows the measurement of the substrate resistivity.

A four probe apparatus consists of four probes equally spaced (typically of the order of S=1 mm). The apparatus is placed on a sample such as a thin semicoductor whose sheet resistance is required to be measured. The figure shows four probes, 1, 2, 3, and 4. A current is passed through the outer probes, 1, and 4. The voltage developed across the two inner probes, 2 and 3 is measured. The sheet resistance is given by,

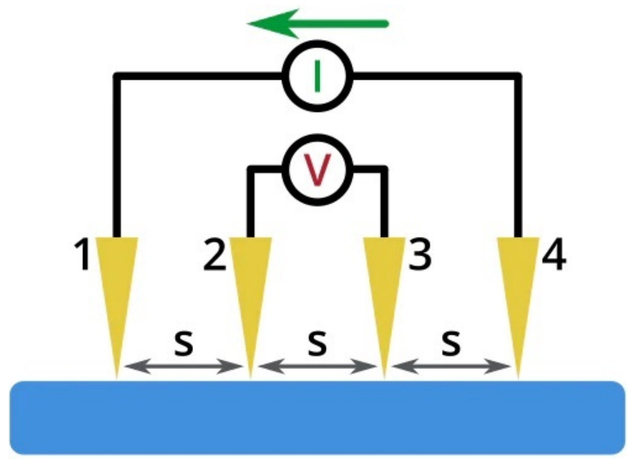
$$R_S = \frac{\pi}{\ln 2} \frac{\Delta V}{I}$$

where,

- R_S = sheet resistance in $\Omega/\Box$
- ΔV = voltage measured across the inner probes
- I = current passed through the outer probes

Applications of four probe nethod

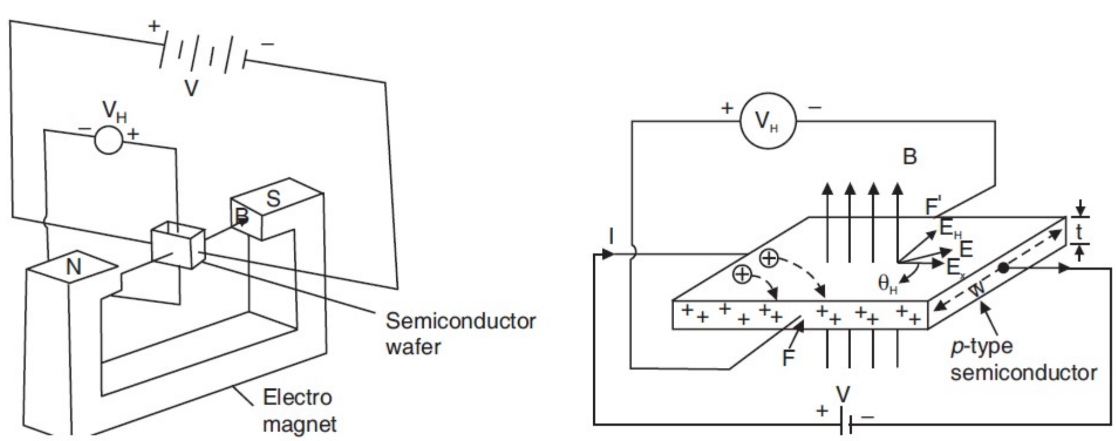
- **Resistance mesurement in thin semiconducting layers**
- **Band gap mesaurements**



8 Hall effect and Hall coefficient

In a material if an electric and magnetic fields are applied perpendicular to each other, then a voltage is developed across the material perpendicular to both the electric and magnetic field directions.This is known as Hall effect. The developed voltage is known as Hall voltage.

Let us consider a p-type semiconductor. Let a potential difference V be applied across its ends. A current of strength I flows through it along the x-direction. In a p-type semiconductor, holes are the majority charge carriers. The current through the wafer is given by



$$I = peAv_d$$

where p is the whole concentration, A the area of cross-section of the end face of semiconductor wafer, e is the charge on the hole (-e is the charge on the electron) and v_d is the drift velocity. If a magnetic field B is applied normal to the wafer surface and hence to the direction of current flow in it, then a transverse potential difference is produced between faces F and F'. It is known as Hall Voltage V_H. On application of magnetic field B, the holes experience a sideways deflection due to the magnetic force F_L, which is given by

$$F_L = eBv_d$$

Holes are deflected toward the front face F and accumulate there. Due to this, a corresponding equivalent negative charge is left on the rear face F'. These opposite charges produce a transverse electric field, E_H, whose direction is from the front to the rear face. Due to the action of E_H, holes experience an electric force in addition to the Lorentz force F_L. When the force F_E due to this transverse electric field balances the magnetic force F_L, equilibrium condition is attained and the holes once again flow along x-direction parallel to the faces F and F'. Under these conditions,

$$\begin{aligned} F_E &= F_L \\ eE_H &= ev_dB \end{aligned}$$

If w is the width of the semiconductor wafer, $E_H = V_H$

$$\frac{V_H}{w} = Bv_d$$

Since, $v_d = J_x/pe$, we have

$$\begin{aligned}
\frac{V_H}{w} &= \frac{BJ_x}{pe} \\
V_H &= \frac{BJ_x w}{pe} \\
V_H &= \frac{BIw}{peA}
\end{aligned}$$

If t is the thickness of the semiconductor plate, $A = wt$. Hence,

$$\begin{aligned}
V_H &= \frac{BIw}{pewt} \\
V_H &= \frac{BI}{pet}
\end{aligned}$$

This is the expression for the Hall voltage V_H.

Hall coefficient:

$$R_H = \frac{E_H}{J_x B} \tag{4}$$

Hall coefficient, R_H is defined as the Hall field per unit current density per unit magnetic induction.

Using $V_H = BI/pet$, $E_H = V_H/w$, we get,

$$\begin{aligned}
R_H &= \frac{BJ_x}{peJ_x B} \\
R_H &= \frac{1}{pe}
\end{aligned}$$

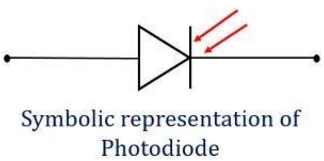
In the above derivation, if the semiconductor is n-type, replace p by n. The direction of the electric fileld will be opposite and the Hall coefficient will be negative. The above equation is derived assuming that the p-semiconductor contains only holes. When the two types of charge carriers are taken into account, it can be shown that the Hall coefficient is given by

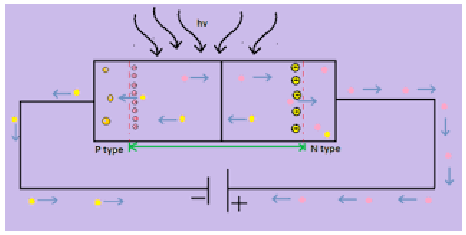
$$R_H = \frac{p\mu_h^2 - n\mu_e^2}{e(p\mu_h + n\mu_e)^2} \tag{5}$$

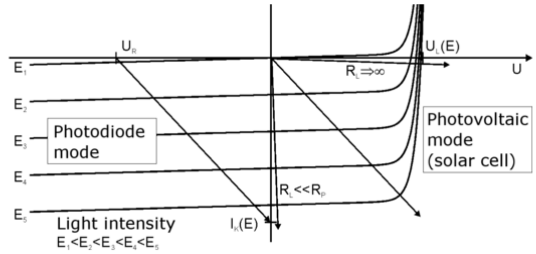
where μ_h, μ_e are the hole and electron mobilities, respectively. The sign of the hall voltage is positive.If it is an n-type semiconductor, the hall voltage will be negative. Knowing the sign of the hall voltage, one can determine whether it is an n-type semiconductor or a p-type semiconductor.

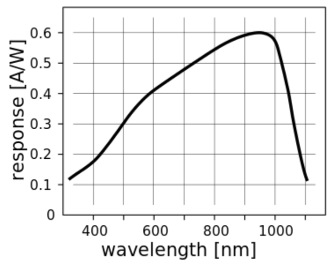
Applications of Hall Effect:

- **Determination of the type of semiconductor, p-type or n-type**
- **Determination of the carrier mobility**
- **Determination of the carrier concentration**
- **Determination of the magnetic flux density**









9 Photodiode

A photodiode is a type of diode, that is, a PN junction that converts light energy into electrical energy when incident light falls on it. The magnitude of the current is directly proportional to the intensity of the incident light. The photodiode is designed to work in reverse bias. If the intensity of light at the PN junction of the photodiode is increased the reverse current also increases in the photodiode. The photodiode has a transparent window with a lens system to collect light into the PN junction area. The response of the photodiode depends on the wavelength of the light and is shown in the figure. The photodiode can be used in photovoltaic mode or photodiode mode.

1. **Photovoltaic mode:** In this mode, when light falls on the junction, electron-hole pairs are created. The photodiode is zero biased. Because of the creation of electron-hle pairs, a voltage is developed across the terminals. This is the basis on which a solar cell works. A solar cell is a large area photodiode.

2. **Photoconductive mode:** In this mode, the photodiode is reverse biased. In the reverse bias mode, when light falls on the junction, the generated electron-hole pairs are separated out and flow out of the junction. The current is proportional to the intensity of the incident light. This can be used as a light detector and smoke detector.

Photodiode characteristics:
The quantum efficiency of a photodiode (photodetector) is given by

$$\begin{aligned} \eta &= \frac{\text{number of electron-hole pairs generated}}{\text{number of incident photons}} \\ &= \frac{I/e}{P_0} \end{aligned} \tag{6}$$

where,

- I = current
- e = electronic charge
- P_0 = incident optical power

The performance of a photodiode is characterized by the responsivity R, which specifies the photocurrent generated per unit optical power. This is given by

$$R = \frac{I_L}{P_0} \tag{7}$$

where T_L is the output photocurrent in amperes and P_0 is the incident optical power in watts. It can be shown that the responsivity R is related to quantum efficiency through the following relation.

$$\begin{aligned} R &= \frac{\eta e}{h\nu} \\ &= \frac{n\lambda}{1.24}\mu m \end{aligned}$$

It is seen that the responsivity is directly proportional to the quantum efficiency at a particular wavelength. The maximum photocurrent in a photodiode equals

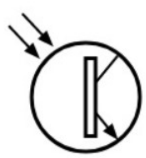
$$I_{L(max)} = \frac{e}{h\nu}P_0$$

Applications of photodiodes

- **Light meters in cameras**
- **Smoke detectors**
- **Barcode scanner**
- **Optical remote control**

10 Phototransistor

A Phototransistor is an electronic switching and current amplification component which relies on exposure to light to operate. When light falls on the junction, reverse current flows which is proportional to the luminance (light intensity). A phototransistor is a bi-polar transistor in which the base region is exposed to light. It is available



Phototransistor Symbol

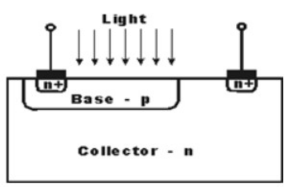
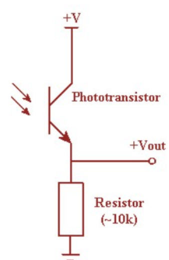
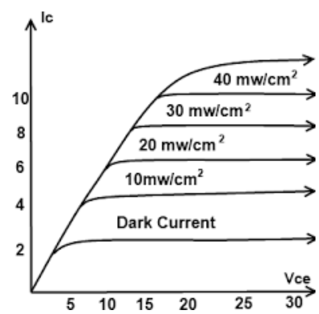


Photo Transistor Construction



Phototransistor Circuit Diagram



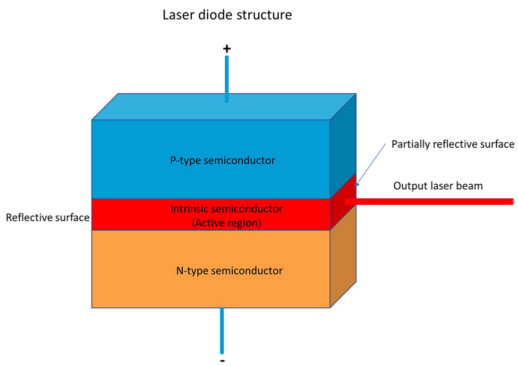
in both the P-N-P and N-P-N types having different configurations like common emitter, common collector, and common base but generally, common emitter configuration is used. It can also work while the base is made open. Compared to the conventional transistor it has more base and collector areas.

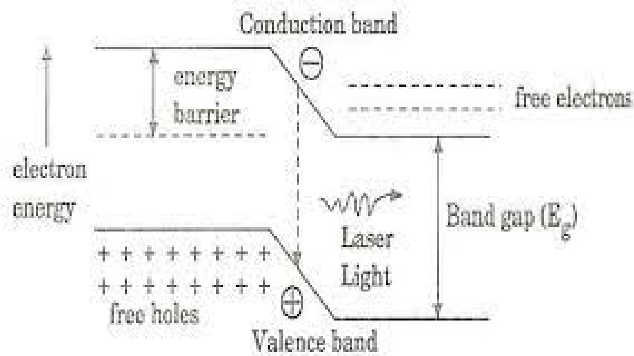
With light falling on the collector-base junction the current flow increases. With the base connection open circuit, the collector-base current must flow in the base-emitter circuit, and hence the current flowing is amplified by normal transistor action. When light falls on the base region, electron-hole pairs are created.

The collector-base junction is very sensitive to light. When light falls on the base region, electron-hole pairs are created. Its working condition depends upon the intensity of light. The base current from the incident photons is amplified by the gain of the transistor, resulting in current gains that range from hundreds to several thousand. A phototransistor is 50 to 100 times more sensitive than a photodiode with a lower level of noise.

Applications of phototransistors:

- **Used in Level indicators**
- **In counting systems, such as currency counters and coin counters**
- **Used in alarm Systems**





11 Semiconductor diode laser

A semiconductor diode laser is a type of p-n junction diode. When the p-n junction diode is forward biased, the electrons from n − region and the holes from the p- region cross the junction and recombine with each other. During the recombination process, the light radiation (photon) is released from a certain specified direct band gap semiconductors like Ga-As. This light radiation is known as recombination radiation. The photon emitted during recombination stimulates other electrons and holes to recombine. As a result, stimulated emission takes place which produces laser.

A laser diode consists of an intrinsic layer of semiconductor between an n-type and a p-type layer as shown. One end perpendicular to the layers is completely polished and the other layer is made partially reflecting. When a forward bias is applied to the diode, the depletion region narrows and the energy levels between the p- and n- layers get modified as shown. Because of the bias, electrons cross the junction and exciting the electrons to jump from the valence band to the conduction band. The electrons return to the valence band causing the emission of a spontaneous photon (in the case of direct band gap semiconductors). The photon will induce further stimulated emission of photons when it strikes an excited atom. The photons travel between the two faces causing a large number of stimulated emissions, which comes out as a laser beam through the partially reflecting face. The light is usually divergent. This is converted to a narrow parallel beam using a lens.

Applications:

- Laser pointer
- Used in CD/ DVD players
- Used in optical fiber communication
- Bar code readers

12 Numerical Problems on Hall effect and electrical conductivity

1. The following data are given for intrinsic germanium at 300 K: $n_i = 2.4 \times 10^{19}/m^3, \mu_e = 0.39\ m^2v^{-1}\ s^{-1}, \mu_h = 0.19\ m^2v^{-1}s^{-1}$. Calculate the resistivity of the sample.

   Solution:

   The conductivity $\sigma = n_i e(\mu_e + \mu_h) = 2.4 \times 10^{19} \times 1.6 \times 10^{-19}(0.39 + 0.19) = 3.84 \times 0.58 = 2.2272$
   The resistivity $\rho = 1/\sigma = 1/2.2272 = 0.448\ \Omega.m$

2. For intrinsic galium-arsenide (GaAs), the room temperature electrical conductivity is $10^{-6}\ /\Omega.m$, the electron and hole mobilities are respectively 0.85 $m^2/V.s$ and 0.04 $m^2/V.s$. Compute the intrinsic carrier concentration at room temperature.

   Solution:

$$\begin{aligned}
\sigma &= n_i e(\mu_e + \mu_h) \\
n_i &= \frac{\sigma}{\mu_e + \mu_h} \\
&= \frac{10^{-6}}{1.6 \times 10^{-19} \times (0.85 + 0.04)} \\
&= 7.0 \times 10^{12}/m^3
\end{aligned}$$

3. An n-type germnanium sample has a donor density of $10^{21}/m^3$. It is arranged in a Hall experiment having magnetic field of 0.5 T and the current density is 500 A/m^2. Find the Hall voltage if the sample is 3 mm wide.

   Solution: The Hall voltage is given by,

$$\begin{aligned}
V_H &= \frac{BJw}{ne} \\
&= \frac{0.5 \times 500 \times 3 \times 10^{-3}}{10^{21} \times 1.6 \times 10^{-19}} \\
&= 4.7 \times 10^{-3}\ V
\end{aligned} \tag{8}$$

4. A copper strip 2.0 cm wide and 1.0 mm thick is placed in a magnetic field with B=1.5 wb/m^2. If a current of 200 A is set up in the strip, calculate Hall voltage that appears across the strip. Assume $R_H = 6 \times 10^{-7}\ m^3/C$.

   Solution:

   The Hall coefficient is given by,

$$R_H = \frac{E_H}{JB}$$

   The sample is of width 2 cm = 2×10^{-2} m and thickness 1 mm = 1×10^{-3} m. Hence, the current density $J = I/A$ can be calculated as

$$J = I/A = 200/(2 \times 10^{-2} \times 1 \times 10^{-3}) = 10^7\ A/m^2$$

   Since, the Hall voltage is $V_H = E_H \times$the width

$$V_H = E_H \times 2 \times 10^2\ (V)$$

   Expressing E_H as $E_H = R_H JB$, we have,

$$\begin{aligned}
V_H &= E_H t \\
V_H &= R_H JBt \\
&= 6 \times 10^{-7} \times 10^7 \times 1.5 \times 2 \times 10^{-2}\ V \\
&= 0.18\ V
\end{aligned} \tag{9}$$